A Weighted Integral Method for Steady State Stokes and Navier-Stokes Equations

Tags: #CFD #Differential_Equations #Physics

Note

Let us reformulate the problem of Stokes flow of Lid-Driven Cavity with an auxiliary function ϕ such that

$$\begin{cases} \nabla^2\omega = 0 \\ \nabla^2\psi = \omega \\ \nabla^2\phi = 1, \quad \phi_{BC} = 0 \end{cases}$$

where ϕ_{BC} is the Dirichlet condition of ϕ. Applying the Green identity, we have the following expression:

$$\int_{\Omega} (\phi\nabla^2\omega - \omega\nabla^2\phi)\, d\Omega = \oint_{\partial\Omega} (\phi\frac{\partial\omega}{\partial n} - \omega\frac{\partial\phi}{\partial n})\, ds$$

where $\Omega \subset \mathbb{R}^2$ and $\partial\Omega$ is the boundary. Substituting the Poisson equation of ϕ and the Dirichlet condition into this relation, we have

$$\int_{\Omega} \omega\, d\Omega = \oint_{\partial\Omega} \frac{\partial\psi}{\partial n}\, ds = \oint_{\partial\Omega} \omega\frac{\partial\phi}{\partial n}\, ds$$

If we use that relation in numerical scheme, then we will obtain

$$\oint_{\partial\Omega} (\frac{\partial\tilde{\psi}^{(i+1)}}{\partial n} - \frac{\partial\tilde{\psi}^{(i)}}{\partial n})\, ds = \oint_{\partial\Omega} (\tilde{\omega}^{(i+1)} - \tilde{\omega}^{(i)})\frac{\partial\phi}{\partial n}\, ds$$

Assuming that the exact solution is reached at $i+1$, then we can set

$$\frac{\partial\tilde{\psi}^{(i+1)}}{\partial n}|_{\partial\Omega} = \frac{\partial\psi_{exact}}{\partial n}|_{\partial\Omega}$$

where $\frac{\partial\psi_{exact}}{\partial n}|_{\partial\Omega}$ is a prescribed boundary condition of the problem.

Numerical Scheme for Stokes flow

On each boundary point, we propose a point-wise relation

$$(\tilde{\omega}_j^{(i+1)} - \tilde{\omega}_j^{(i)})(\frac{\partial \phi}{\partial n})_j = (\frac{\partial \psi_{exact}}{\partial n})_j - (\frac{\partial \tilde{\psi}^{(i)}}{\partial n})_j$$

and the prototype of boundary vorticity value update scheme can be formulated as

$$\tilde{\omega}_j^{(i+1)} = \tilde{\omega}_j^{(i)} + (\frac{\partial \phi}{\partial n})_j^{-1}[(\frac{\partial \psi_{exact}}{\partial n})_j - (\frac{\partial \tilde{\psi}^{(n)}}{\partial n})_j]$$

The numerical scheme can be written as the following steps:

1. Give the initial guess of boundary vorticity value and compute the normal gradient of ϕ:

$$\tilde{\omega}_{BC}^{(n)} = \tilde{\omega}_{BC}^{(0)}\,, \quad \frac{\partial \phi}{\partial n}$$

2. Solve the Laplace equation

$$\nabla^2 \tilde{\omega} = 0$$

3. Solve the Poisson equation

$$\nabla^2 \tilde{\psi} = \tilde{\omega}$$

4. Calculate the normal gradient

$$\frac{\tilde{\psi}^{(n)}}{\partial n}$$

5. Update the boundary vorticity value with the expression:

$$\tilde{\omega}_j^{(i+1)} = \tilde{\omega}_j^{(i)} + \alpha(\frac{\partial \phi}{\partial n})_j^{-1}[(\frac{\partial \psi_{exact}}{\partial n})_j - (\frac{\partial \tilde{\psi}^{(i)}}{\partial n})_j]$$

   where α is a relaxation coefficient.

Remark:

- The normal gradient of ϕ only needs to be computed once, since it only depends on the geometry of the domain Ω.

Thought

- This method provides us a very interesting insight about the boundary vorticity of Stokes flow. According to its formulation, this method is based on the distribution analysis of the source term of the vorticity equation, which is a bit trivial (or degenerating) since this source term is zero in the domain, i.e

$$\int_\Omega \phi \nabla^2 \omega \, d\Omega = \int_\Omega \phi \, f \, d\Omega = 0$$

However, with this projection, we can immediately observe that the distribution of boundary vorticity depends only on the Neumman condition of the problem

$$\oint_{\partial\Omega} \frac{\partial \psi}{\partial n}\, ds = \oint_{\partial\Omega} \omega \frac{\partial \phi}{\partial n}\, ds$$

where ϕ can be considered as a geometrical weight as it depends only on the geometry of the domain. However, we do not suggest considering it as a physical interpretation but merely as a numerical coincidence. Comparing with the integral constraint given by the Quartapelle's projection condition

$$\int_{\Omega} \omega\eta\, d\Omega = \oint_{\partial\Omega} (\frac{\partial \psi}{\partial n}\eta - \psi\frac{\partial \eta}{\partial n})\, ds$$

where η is an arbitrary harmonic function in the domain, we may conclude that the distribution of the boundary vorticity for a **Stokes flow** does not require the information of Dirichlet condition of the problem.

- However, the point-wise relation is not clarified, because the integral equivalence does not imply a point-wise equivalence. For example, considering the integrals of two functions $f(x)$ and $g(x)$:

$$I_1 = \int_0^1 f(x)\, dx = \int_0^1 (x^2 + \frac{2}{3})\, dx$$

and

$$I_2 = \int_0^1 g(x)\, dx = \int_0^1 (x^3 + \frac{3}{2}x)\, dx$$

We have

$$I_1 = I_2 = 1$$

but it does not imply that $f(x) = g(x), \forall x \in [0, 1]$. Actually, this method lacks of some extra constraints to guarantee the solution uniqueness, because what we have is

$$\sum^{N_{BC}} (\frac{\partial \psi}{\partial n})_i \Delta = \sum^{N_{BC}} \omega_i (\frac{\partial \phi}{\partial n})_i \Delta$$

and there is no local constraint on each boundary point.

Info

According to the simulation result, in the Lid-Driven Cavity of a Stokes flow, there is a numerical point-wise equivalence on the boundary, but we cannot conclude whether the result is an algebraic equivalence (numerical coincidence) or it is dominated by the nature of the differential operator.

Question

Can we extend this method to the problem of steady state Navier-Stokes equation ?

Let us recall the stream function-vorticity formulation of the steady state Navier-Stokes equation with boundary conditions

$$\begin{cases} \nabla^2\psi = \omega \\ \nabla^2\omega - Re\, g(\psi,\omega) = 0 \\ \psi|_\Omega = a \\ \dfrac{\partial\psi}{\partial n}|_\Omega = b \end{cases}$$

where a and b could be constants or scalar functions, and

$$g(\psi,\omega) = (\mathbf{u}\cdot\nabla)\omega = -\frac{\partial\psi}{\partial y}\frac{\partial\omega}{\partial x} + \frac{\partial\psi}{\partial x}\frac{\partial\omega}{\partial y}$$

with the convention

$$u = -\frac{\partial\psi}{\partial y} \qquad v = \frac{\partial\psi}{\partial x}$$

Applying the Green Identity, we have

$$\int_\Omega (\phi\nabla^2\omega - \omega\nabla^2\phi)\, d\Omega = \oint_{\partial\Omega} (\phi\frac{\partial\omega}{\partial n} - \omega\frac{\partial\phi}{\partial n})\, ds$$

Since $\phi_{BC} = 0$, we have

$$\int_\Omega \phi\nabla^2\omega\, d\Omega = Re\int_\Omega \phi f(\psi,\omega)\, d\Omega = -\oint_{\partial\Omega} \omega\frac{\partial\phi}{\partial n}\, ds + \int_\Omega \omega\, d\Omega$$

Using the definition of circulation

$$\int_\Omega \omega\, d\Omega = \oint_{\partial\Omega} \frac{\partial\psi}{\partial n}\, ds$$

eventually we obtain

$$Re\int_\Omega \phi g(\psi,\omega)\, d\Omega = \oint_{\partial\Omega} \frac{\partial\psi}{\partial n}\, ds - \oint_{\partial\Omega} \omega\frac{\partial\phi}{\partial n}\, ds$$

ϕ is unique and can be determined analytically or numerically (depending on the geometry of the domain), however this projection cannot provide the exact form of distribution of boundary vorticity value because it depends on the convection term, which is considered as unknown, in the entire domain. Thus, the current test function

$$\nabla^2\phi = 1, \quad \phi|_{\partial\Omega} = 0$$

is not practical in the resolution of steady state Navier-Stokes equation.

Question

What will happen if we replace the previous auxiliary equation of ϕ with

$$\nabla^2\phi + Re\nabla\cdot(u_0\phi) = 1$$
$$\phi|_{\partial\Omega} = 0$$

The PDE system formed by the Navier-Stokes equation (with frozen velocity) and the auxiliary equation can be written as

$$\begin{cases} \nabla^2\psi = \omega \\ \nabla^2\omega - Re\nabla\cdot(u_0\omega) = 0 \\ \nabla^2\phi + Re\nabla\cdot(u_0\phi) = 1 \end{cases}$$

where

$$\phi|_{\partial\Omega} = 0$$

The weighted integral of the steady state vorticity transport equation is

$$\int_\Omega \phi(\nabla^2\omega - Re\nabla\cdot(u_0\omega))\,d\Omega = 0$$

By the second Green's identity, we have

$$\int_\Omega \phi\nabla^2\omega\,d\Omega = \int_\Omega \omega\nabla^2\phi\,d\Omega + \int_{\partial\Omega}\phi\frac{\partial\omega}{\partial n}\,d\Gamma - \int_{\partial\Omega}\omega\frac{\partial\phi}{\partial n}\,d\Gamma$$

and

$$\begin{aligned} Re\int_\Omega \phi\nabla\cdot(u_0\,\omega)\,d\Omega &= Re\int_\Omega \nabla\cdot(\phi\,\omega\,u_0)\,d\Omega - Re\int_\Omega \omega\,u_0\cdot\nabla\phi\,d\Omega \\ &= Re\int_{\partial\Omega}\phi\,\omega\,u_0\cdot\mathbf{n}\,d\Gamma - Re\int_\Omega \omega\,u_0\cdot\nabla\phi\,d\Omega \end{aligned}$$

Substituting the Dirichlet condition of ϕ into the integrals, we have

$$\begin{aligned} \int_\Omega \phi(\nabla^2\omega - Re\nabla\cdot(u_0\omega))\,d\Omega &= \int_\Omega \omega\nabla^2\phi\,d\Omega - \int_{\partial\Omega}\omega\frac{\partial\phi}{\partial n}\,d\Gamma - Re\int_\Omega \omega\,u_0\cdot\nabla\phi\,d\Omega \\ &= \int_\Omega \omega(\nabla^2\phi + Re\nabla\cdot(u_0\phi))\,d\Omega - \int_{\partial\Omega}\omega\frac{\partial\phi}{\partial n}\,d\Gamma \end{aligned}$$

Thus

$$\int_\Omega \omega\,d\Omega = \int_{\partial\Omega}\frac{\partial\psi}{\partial n}\,d\Gamma = \int_{\partial\Omega}\omega\frac{\partial\phi}{\partial n}\,d\Gamma$$

Observation

We can observe that the advection term in the equation of ϕ is opposite to the advection term of the vorticity transport equation. Actually the differential operator applied to ϕ is an *adjoint operator* of the differential operator applied to the vorticity.

A possible numerical scheme

Let the prescribed boundary velocity be denoted by

$$\mathbf{u}_\Gamma \cdot \mathbf{t} = \frac{\partial \psi}{\partial n}$$

where $\mathbf{t}$ is the tangent vector along the boundary. We consider a correction relation such that

$$\int_{\partial\Omega} \frac{\partial \tilde{\psi}^{(i+1)}}{\partial n} - \frac{\partial \tilde{\psi}^{(i)}}{\partial n}\, d\Gamma = \int_{\partial\Omega} (\omega^{(i+1)} - \omega^{(i)}) \frac{\partial \phi^{(i)}}{\partial n}\, d\Gamma$$

For now, we assume that the point-wise relation is valid, then we will obtain

$$\frac{\partial \tilde{\psi}^{(i+1)}}{\partial n} - \frac{\partial \tilde{\psi}^{(i)}}{\partial n} = (\omega_{BC}^{(i+1)} - \omega_{BC}^{(i)}) \frac{\partial \phi^{(i)}}{\partial n}$$

$\phi^{(i)}$ is a solution satisfying

$$\nabla^2 \phi^{(i)} + Re \nabla \cdot (\mathbf{u}_0^{(i)} \phi^{(i)}) = 1$$

where

$$\mathbf{u}_0^{(i)} = -\nabla \times \tilde{\psi}^{(i-1)}$$

Assuming that

$$\frac{\partial \tilde{\psi}^{(i+1)}}{\partial n} = \frac{\partial \psi_{exact}}{\partial n}$$

thus the update scheme of the boundary vorticity is

$$\omega_{BC}^{(i+1)} = \omega_{BC}^{(i)} + \alpha \left(\frac{\partial \phi^{(i)}}{\partial n}\right)^{-1} \left(\frac{\partial \psi_{exact}}{\partial n} - \frac{\partial \tilde{\psi}^{(i)}}{\partial n}\right)$$

where α is a relaxation number. Therefore, the numerical scheme can be summarized as the following steps:

1. Initialize the flow with Stokes solver.
2. Use the time marching scheme to solve the vorticity transport equation with the frozen velocity given by the stream function in the previous iteration. Solve the Poisson equation of the stream function ψ. Compute its normal derivative.
3. Use the pseudo time marching scheme to solve the convection-diffusion equation of ϕ with the frozen velocity . Calculate its normal derivative on the boundary, then update the boundary vorticity. Iterate this step till the boundary error is small enough.
4. Update the frozen velocity, and redo the step 2 till the boundary error is small.

Reference:

- Quartapelle, L., and Valz-Gris, F. (1981). Projection conditions on the vorticity in viscous incompressible flows. Int. J. Num. Meth. FI. 1, 129.